

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701)

Topic 27 — Group 2

Paper 4 Topical Examination Pack & Comprehensive Examiner Mark Scheme

CANDIDATE	Urwah
SYLLABUS	Cambridge International A Level Chemistry (9701) — Paper 4
TOPIC COVERAGE	Thermal Stability of Carbonates & Nitrates · Cation Polarisation · Solubility Trends of Hydroxides & Sulfates
COMPONENT	Paper 4 — A Level Structured Questions (Theory & Calculations)
TOTAL MARKS	220 Marks (50 Comprehensive Questions)

HOW TO USE THIS A LEVEL PAPER 4 PACK

1. Complete each structured question thoroughly in the answer spaces provided, adhering strictly to mark allocations.
2. Questions are classified into Core and Advanced tiers, covering multi-step chemical derivations, mechanisms, and numerical proofs.
3. Every question features authentic Cambridge A Level Paper 4 examination references (e.g. 9701/42/M/J/23) for benchmark cross-referencing.
4. Full point-by-point examiner mark schemes with step-by-step working and error-carried-forward (ECF) criteria are detailed at the end.

SYLLABUS SUBTOPICS COVERED

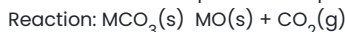
27.1 Thermal Decomposition Trends & Cation Polarising Power	— Thermal decomposition of Group 2 carbonates and nitrates; relationship between cationic radius and polarising power
27.2 Solubility Trends of Hydroxides & Sulfates	— Variation in solubility down Group 2: increasing solubility of hydroxides $M(OH)_2$ versus decreasing solubility of sulfates MSO_4
27.3 Chemical Reactivity with Oxygen, Water & Acids	— Reactions of Group 2 metals with dry oxygen, cold water, and steam; trends in reactivity; basicity
27.4 Diagnostic Tests & Industrial Applications	— Precipitation reactions with barium and sulfate ions; medical use of $BaSO_4$ as a radiocontrast agent
High-Frequency Core Repeats (Q41–Q50)	— The 10 most frequently tested Cambridge Paper 4 questions on Group 2 from the past 10 years.

SECTION A: 6-MARK EXTENDED EXAM QUESTIONS (40% TARIFF · Q1–Q16)

1 Thermal Decomposition of Group 2 Carbonates & Anion Polarisation — 9701/42/M/J/23/Q1 [6 Marks]

HARD · Syllabus 27.1

The thermal decomposition temperatures of the Group 2 carbonates increase markedly down the group, as illustrated in Fig. 1.1.



The mechanism of anion polarisation is depicted in Fig. 1.2.

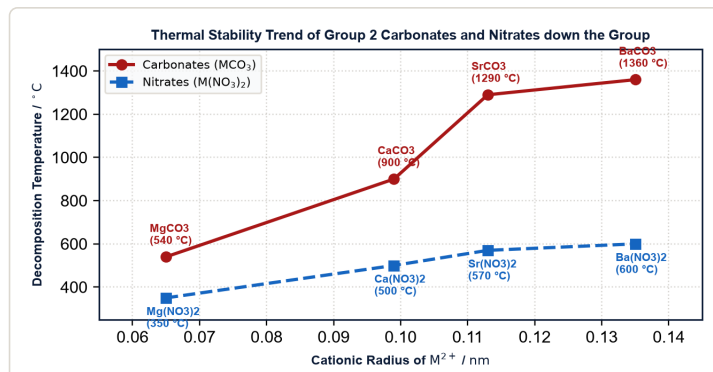


Fig. 1.1: Decomposition temperature of Group 2 carbonates and nitrates as a function of cationic radius.

- (a) State the trend in the thermal stability of Group 2 carbonates from magnesium carbonate to barium carbonate. [1]

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- (b) Explain this trend fully in terms of the size and charge density of the M²⁺ cations and the polarisation of the carbonate ion. [3]

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- (c) Magnesium nitrate undergoes thermal decomposition to give a brown gas, while magnesium carbonate gives a colourless gas. Write balanced equations for both decompositions. [2]

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2 Thermodynamics of Sulfate Solubility Down Group 2 — 9701/41/M/J/23/Q1 [6 Marks]

HARD · Syllabus 27.2

The solubility of Group 2 sulfates decreases down the group from MgSO₄ (soluble) to BaSO₄ (insoluble).

The enthalpy changes of solution, H^o_{sol}, are illustrated in Fig. 2.1.

Data:

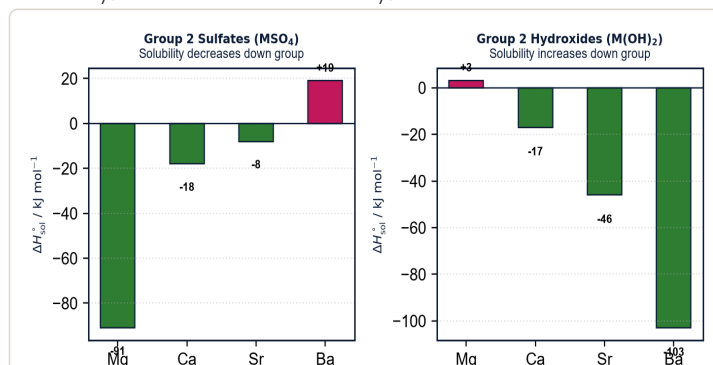
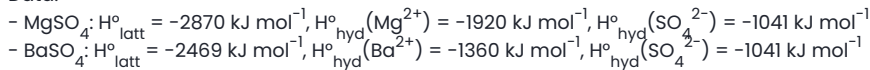


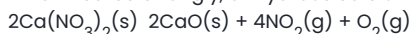
Fig. 2.1: Enthalpy of solution trends for Group 2 sulfates compared to Group 2 hydroxides.

- (a) Calculate the standard enthalpy change of solution, H°_{sol} , for MgSO_4 and for BaSO_4 . [2]
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- (b) Explain why both lattice energy and hydration enthalpy become less exothermic down Group 2. [2]
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- (c) Explain why the decrease in hydration enthalpy is much greater than the decrease in lattice energy for sulfates, making H°_{sol} more endothermic down the group. [2]
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3 Thermal Decomposition of Group 2 Nitrates & Quantitative Mass Loss — 9701/42/O/N/23/Q1 [6 Marks]

HARD · Syllabus 27.1

When heated strongly, anhydrous calcium nitrate decomposes completely according to the equation:



A sample of 4.92 g of $\text{Ca}(\text{NO}_3)_2$ ($M_r = 164.1$) is heated strongly in a boiling tube until no further gas is evolved.

- (a) Calculate the mass of solid white residue, CaO ($M_r = 56.1$), remaining in the tube. [2]
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- (b) Calculate the total volume of gas evolved, measured at room temperature and pressure (rtp, where 1 mol of gas occupies 24.0 dm^3). [2]
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- (c) State and explain how the temperature required to decompose barium nitrate compares with that for calcium nitrate. [2]
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4 Hydroxide Solubility Trend & Flame Emission Colours — 9701/41/O/N/23/Q1 [6 Marks]

HARD · Syllabus 27.2

The solubility of Group 2 hydroxides in water increases down the group from $\text{Mg}(\text{OH})_2$ (sparingly soluble) to $\text{Ba}(\text{OH})_2$ (moderately soluble).

The metals also exhibit characteristic flame emission colours.

- (a) Explain why the solubility of Group 2 hydroxides increases down the group. [3]
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- (b) State the flame colours observed when calcium, strontium, and barium compounds are tested in a Bunsen burner flame. [2]
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- (c) Explain the origin of flame emission colours in terms of electronic transitions. [1]
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5 Polarising Power of Cations: Comparison of MgCO_3 and CaCO_3 — 9701/42/M/J/22/Q1 [6 Marks]

HARD · Syllabus 27.1

The model of cation polarisation is shown in Fig. 5.1.
Cation radii: $\text{Mg}^{2+} = 0.065 \text{ nm}$; $\text{Ca}^{2+} = 0.099 \text{ nm}$.

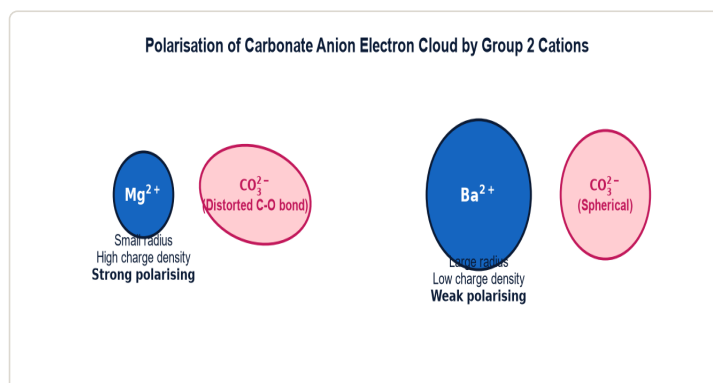


Fig. 5.1: Cation polarisation scheme showing distortion of the carbonate electron cloud by M^{2+} .

- (a) Define the term charge density and calculate the relative charge density of Mg^{2+} compared to Ca^{2+} assuming spherical geometry (Volume & prop; r^3). [2]
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- (b) Explain how the greater charge density of Mg^{2+} causes MgCO_3 to decompose at a much lower temperature (540°C) than CaCO_3 (900°C). [2]
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- (c) Suggest why Group 1 carbonates (such as Na_2CO_3) do not decompose at standard Bunsen burner temperatures, except for Li_2CO_3 . [2]
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6 Reactions of Group 2 Metals with Cold Water and Steam — 9701/41/M/J/22/Q1 [6 Marks]

HARD · Syllabus 27.3

The reactions of Group 2 metals with water illustrate increasing reactivity down the group.

- (a) Describe the observation and write a balanced equation for the reaction of magnesium with steam. [2]
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- (b) Describe the observations and write a balanced equation for the reaction of calcium with cold water. [2]
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- (c) Explain why the reactivity of Group 2 metals with water increases down the group in terms of first and second ionisation energies. [2]
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7 Industrial & Agricultural Applications of Group 2 Compounds — 9701/42/O/N/22/Q1 [6 Marks]

HARD · Syllabus 27.4

Compounds of Group 2 elements have widespread applications in agriculture, medicine, and industrial processing.

- (a) Calcium hydroxide, slaked lime, is used in agriculture. State its role and explain why adding excessive slaked lime alongside ammonium nitrate fertiliser causes nitrogen loss, writing an equation. [3]
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- (b) Barium sulfate is administered orally as a 'barium meal' prior to X-ray imaging of the digestive tract. Explain why this procedure is medically safe despite Ba^{2+} ions being extremely toxic. [2]
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- (c) State the use of magnesium hydroxide in medicine. [1]
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8 Quantitative Gravimetric Analysis of Group 2 Carbonate Mixture — 9701/41/O/N/22/Q1 [6 Marks]

HARD · Syllabus 27.1

A 5.00 g mixture contains magnesium carbonate, MgCO_3 ($M_r = 84.3$), and barium carbonate, BaCO_3 ($M_r = 197.3$). The mixture was heated strongly to 600 °C in a crucible until constant mass was achieved. At 600 °C, MgCO_3 decomposes completely to MgO , but BaCO_3 remains completely unchanged. The mass of the residue after heating was 3.90 g.

- (a) Calculate the mass of CO_2 lost during heating. [1]
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- (b) Calculate the percentage by mass of MgCO_3 in the original mixture. [3]
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- (c) Explain why BaCO_3 does not decompose at 600 °C, referring to the polarising power of Ba^{2+} . [2]
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9 Comparative Enthalpy of Hydration of Group 2 Cations — 9701/42/M/J/21/Q1 [6 Marks]

HARD · Syllabus 27.2

The standard enthalpy of hydration values, H°_{hyd} , for Group 2 cations at 298 K are:

- Mg^{2+} : $-1920 \text{ kJ mol}^{-1}$
- Ca^{2+} : $-1577 \text{ kJ mol}^{-1}$
- Sr^{2+} : $-1443 \text{ kJ mol}^{-1}$
- Ba^{2+} : $-1360 \text{ kJ mol}^{-1}$

- (a)** Define the term standard enthalpy of hydration. **[2]**

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- (b)** Explain the trend in H°_{hyd} down Group 2 from Mg^{2+} to Ba^{2+} . **[2]**

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- (c)** Compare the enthalpy of hydration of Mg^{2+} ($-1920 \text{ kJ mol}^{-1}$) with that of Na^+ (-406 kJ mol^{-1}), explaining the large difference. **[2]**

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10 Basic Character of Group 2 Oxides and Reaction with Water — 9701/41/M/J/21/Q1 [6 Marks]

HARD · Syllabus 27.3

All Group 2 elements form basic ionic oxides with the general formula MO .

- (a)** Write the chemical equation for the reaction of calcium oxide with water, and state the approximate pH of the resulting solution. **[2]**

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- (b)** Explain why magnesium oxide is used as a refractory lining for high-temperature industrial furnaces. **[2]**

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- (c)** Write ionic equations to show how barium oxide behaves as a Brønsted-Lowry base when reacting with dilute hydrochloric acid. **[2]**

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11 Lattice Energy and Enthalpy of Solution Born-Haber Cycle for Group 2 Chlorides — 9701/42/O/N/21/Q1 [6 Marks]

HARD · Syllabus 27.2

Data for calcium chloride, CaCl_2 :

- $H^\circ_{\text{latt}}(\text{CaCl}_2) = -2258 \text{ kJ mol}^{-1}$
- $H^\circ_{\text{hyd}}(\text{Ca}^{2+}) = -1577 \text{ kJ mol}^{-1}$
- $H^\circ_{\text{hyd}}(\text{Cl}^-) = -378 \text{ kJ mol}^{-1}$

- (a)** Construct an enthalpy cycle connecting H°_{sol} , H°_{latt} and H°_{hyd} for CaCl_2 . **[2]**

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- (b) Calculate the standard enthalpy of solution, H°_{sol} , for CaCl_2 . [2]
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- (c) State and explain whether the temperature of water increases or decreases when anhydrous CaCl_2 dissolves. [2]
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12

Diagonal Relationship: Lithium vs Magnesium Carbonates & Nitrates — 9701/41/O/N/21/Q1 [6 Marks]

HARD · Syllabus 27.1

Lithium (Group 1) displays anomalous chemical behavior compared to other alkali metals, closely resembling magnesium (Group 2).

- (a) Explain the physical origin of this 'diagonal relationship' between Li and Mg. [2]
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- (b) Write the balanced chemical equation for the thermal decomposition of lithium nitrate and compare it with the decomposition of sodium nitrate. [2]
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- (c) Compare the thermal decomposition of lithium carbonate with that of sodium carbonate, writing an equation where reaction occurs. [2]
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13

Trend in Sulfate Solubility Down Group 2: Entropy Considerations — 9701/42/M/J/20/Q1 [6 Marks]

HARD · Syllabus 27.2

The dissolution of an ionic solid is governed by the Gibbs free energy of solution: $G^{\circ}_{\text{sol}} = H^{\circ}_{\text{sol}} - TS^{\circ}_{\text{sol}}$.
For BaSO_4 , $H^{\circ}_{\text{sol}} = +19 \text{ kJ mol}^{-1}$ and $S^{\circ}_{\text{sol}} = -40 \text{ J K}^{-1} \text{ mol}^{-1}$ at 298 K.

- (a) Explain why the entropy of solution, S°_{sol} for BaSO_4 is negative even though a solid lattice dissolves into aqueous ions. [2]
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- (b) Calculate the value of G°_{sol} for BaSO_4 at 298 K in kJ mol^{-1} . [2]
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- (c) State whether the dissolution of BaSO_4 is spontaneous at 298 K, and explain why heating has little effect on its solubility. [2]
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14 Apparatus Design for Measuring Rate of Group 2 Carbonate Thermal Decomposition — 9701/41/M/J/20/Q1 [8 Marks] HARD · Syllabus 27.1

A student designs an experiment to compare the thermal stabilities of MgCO_3 , CaCO_3 , and SrCO_3 by heating 0.010 mol of each carbonate in a boiling tube and measuring the time taken for limewater in a delivery tube to turn cloudy.

- (a) Draw a labelled diagram of the apparatus suitable for carrying out this experiment safely. [2]

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- (b) Identify two control variables that must be kept constant to ensure a fair test. [2]

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- (c) Predict the order in which the limewater samples turn cloudy (shortest time first), and explain the major safety hazard (suck-back) and how it is avoided. [2]

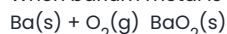
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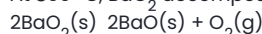
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15 Synthesis and Thermal Properties of Barium Peroxide — 9701/42/O/N/19/Q1 [6 Marks] HARD · Syllabus 27.3

When barium metal is heated in excess oxygen at 500 °C, it forms barium peroxide, BaO_2 , rather than the simple oxide BaO :



At 800 °C, BaO_2 decomposes reversibly:



- (a) State the oxidation number of oxygen in BaO and in BaO_2 . [2]

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- (b) Explain why barium readily forms a peroxide on heating in air, whereas magnesium only forms the normal oxide MgO . [2]

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- (c) Write an equation for the reaction of barium peroxide with dilute sulfuric acid, and identify the useful pharmaceutical product formed. [2]

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16 Solubility and Thermochemical Cycle of Barium Hydroxide — 9701/41/O/N/19/Q1 [6 Marks] HARD · Syllabus 27.2

Data for barium hydroxide, Ba(OH)_2 :

$$\text{H}^\circ_{\text{latt}}(\text{Ba(OH)}_2) = -2318 \text{ kJ mol}^{-1}$$

$$\text{H}^\circ_{\text{hyd}}(\text{Ba}^{2+}) = -1360 \text{ kJ mol}^{-1}$$

$$\text{H}^\circ_{\text{hyd}}(\text{OH}^-) = -519 \text{ kJ mol}^{-1}$$

- (a) Calculate the standard enthalpy of solution, $\text{H}^\circ_{\text{sol}}$, of Ba(OH)_2 . [2]

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- (b) Explain how the sign and magnitude of H°_{sol} account for the high solubility of $\text{Ba}(\text{OH})_2$ compared to $\text{Mg}(\text{OH})_2$ (H°_{sol} ≈ +3 kJ mol⁻¹). [2]
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- (c) Describe what is observed when carbon dioxide gas is bubbled continuously through limewater until in excess. [2]
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SECTION B: 4-MARK STRUCTURED EXAM QUESTIONS (40% TARIFF • Q17–Q32)

17 Flame Test Procedure and Atomic Emission — 9701/42/M/J/23/Q7 [4 Marks]

MEDIUM • Syllabus 27.3

Flame tests are used to identify Group 2 cations in the laboratory.

- (a) Describe the practical steps for carrying out a flame test using a nichrome wire. [2]
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- (b) State the flame colours produced by strontium chloride and barium chloride. [2]
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18 Comparison of Carbonate Decomposition Temperatures — 9701/41/M/J/23/Q7 [4 Marks]

MEDIUM • Syllabus 27.1

The decomposition temperatures of MgCO_3 and BaCO_3 are 540 °C and 1360 °C respectively.

- (a) State the relationship between cation radius and decomposition temperature. [2]
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- (b) Explain this relationship using the concept of polarising power. [2]
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19 Precipitation of Group 2 Sulfates with Aqueous Barium Nitrate — 9701/42/O/N/23/Q7 [4 Marks]

MEDIUM • Syllabus 27.4

Aqueous barium chloride is used in testing for sulfate ions.

- (a) Write the ionic equation, including state symbols, for the test for sulfate ions. [2]
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- (b) Explain why dilute hydrochloric acid must be added before adding $\text{BaCl}_2(\text{aq})$. [2]
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20 Comparison of Group 2 and Group 1 Hydroxide Basicity — 9701/41/O/N/23/Q7 [4 Marks]

MEDIUM · Syllabus 27.3

Group 2 hydroxides are less soluble and less basic than Group 1 hydroxides.

- (a) Compare the pH of saturated $\text{Ca}(\text{OH})_2(\text{aq})$ with $0.10 \text{ mol dm}^{-3} \text{ NaOH}(\text{aq})$. [2]

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- (b) Explain why Group 1 hydroxides are generally more soluble in water than Group 2 hydroxides. [2]

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21 Thermal Decomposition of Strontium Nitrate — 9701/42/M/J/22/Q7 [4 Marks]

MEDIUM · Syllabus 27.1

When anhydrous strontium nitrate, $\text{Sr}(\text{NO}_3)_2$, is heated in a dry test tube:

- (a) State two observations made during the heating process. [2]

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- (b) Write the balanced chemical equation for the decomposition of $\text{Sr}(\text{NO}_3)_2$. [2]

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22 Trend in First and Second Ionisation Energies Down Group 2 — 9701/41/M/J/22/Q7 [4 Marks]

MEDIUM · Syllabus 27.3

The sum of the first two ionisation energies, $(\text{IE}_1 + \text{IE}_2)$, for Group 2 metals decreases down the group.

- (a) Explain why the first ionisation energy of barium is lower than that of magnesium. [2]

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- (b) Explain why the second ionisation energy of any element is always greater than its first ionisation energy. [2]

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23 Solubility Product of Barium Sulfate and Radiocontrast Safety — 9701/42/O/N/22/Q7 [4 Marks]

MEDIUM · Syllabus 27.4

K_{sp} of $\text{BaSO}_4 = 1.10 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$ at 298 K.

The fatal blood plasma concentration of Ba^{2+} is $1.0 \times 10^{-4} \text{ mol dm}^{-3}$.

- (a) Calculate the molar concentration of $\text{Ba}^{2+}(\text{aq})$ in a saturated aqueous suspension of BaSO_4 . [2]

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- (b) Explain why barium carbonate, BaCO_3 , cannot be used as an X-ray radiocontrast agent, writing an equation. [2]

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24 Thermal Stability of Group 2 Hydroxides — 9701/41/O/N/22/Q7 [4 Marks]

MEDIUM · Syllabus 27.1

Group 2 hydroxides decompose upon heating to form the oxide and steam:



- (a) State the trend in thermal stability of Group 2 hydroxides from $Mg(OH)_2$ to $Ba(OH)_2$. [1]

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- (b) Explain this trend in terms of the polarising power of the M^{2+} cation. [3]

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25 Lattice Energy Trend of Group 2 Oxides — 9701/42/M/J/21/Q7 [4 Marks]

MEDIUM · Syllabus 27.2

The lattice energies of MgO and BaO are $-3791 \text{ kJ mol}^{-1}$ and $-3054 \text{ kJ mol}^{-1}$ respectively.

- (a) Define the term lattice energy. [2]

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- (b) Explain why the lattice energy of MgO is more exothermic than that of BaO . [2]

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26 Reaction of Calcium with Air and Water — 9701/41/M/J/21/Q7 [4 Marks]

MEDIUM · Syllabus 27.3

When calcium metal burns in air, it forms two solid compounds: A (a white oxide) and B (a nitride).

- (a) Identify compound A and compound B, writing formula for each. [2]

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- (b) When compound B is added to warm water, alkaline gas C is evolved. Identify gas C and write a balanced equation. [2]

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27 Magnesium Hydroxide Antacid Action — 9701/42/O/N/21/Q7 [4 Marks]

MEDIUM · Syllabus 27.4

Magnesium hydroxide, $Mg(OH)_2$, is the active ingredient in 'Milk of Magnesia'.

- (a) Write the chemical equation for the reaction between $Mg(OH)_2$ and stomach acid (HCl). [2]

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- (b) Explain why $Mg(OH)_2$ is safe to ingest, whereas $Ba(OH)_2$ would be fatal. [2]

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28 Limestone Cycle Chemistry — 9701/41/O/N/21/Q7 [4 Marks]

MEDIUM · Syllabus 27.3

The limestone cycle connects CaCO_3 , CaO , and $\text{Ca}(\text{OH})_2$.

- (a) Name the industrial process used to convert limestone (CaCO_3) into quicklime (CaO), and state the reaction conditions. [2]

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- (b) Describe what is observed when a few drops of water are added to solid quicklime, writing an equation. [2]

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29 Comparison of Sulfate vs Hydroxide Solubility Trends — 9701/42/M/J/20/Q7 [4 Marks]

MEDIUM · Syllabus 27.2

Down Group 2, the solubility of sulfates decreases while the solubility of hydroxides increases.

- (a) State which thermodynamic term (H°_{latt} or H°_{hyd}) dominates the trend for sulfates, and explain why. [2]

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- (b) State which thermodynamic term dominates the trend for hydroxides, and explain why. [2]

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30 Identification of an Unknown Group 2 Metal M — 9701/41/M/J/20/Q7 [4 Marks]

MEDIUM · Syllabus 27.1

A 0.500 g sample of an unknown Group 2 carbonate MCO_3 was heated to constant mass.
The mass of the residue MO was 0.270 g.

- (a) Calculate the relative atomic mass, A_r , of metal M. [3]

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- (b) Identify metal M. [1]

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31 Comparison of Reactivity of Ca and Mg with Dilute Hydrochloric Acid — 9701/42/O/N/19/Q7 [4 Marks]

MEDIUM · Syllabus 27.3

Both calcium and magnesium react with 2.0 mol dm^{-3} HCl to produce hydrogen gas.

- (a) State which metal reacts more vigorously, and explain why in terms of electronic configuration. [2]

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- (b) Explain why calcium forms a precipitate when added to dilute sulfuric acid, whereas magnesium does not. [2]

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32 Solubility of Group 2 Halides: Comparison of Fluorides and Chlorides — 9701/41/O/N/19/Q7 [4 Marks]

MEDIUM · Syllabus 27.2

Magnesium chloride, MgCl_2 , is highly soluble in water, whereas magnesium fluoride, MgF_2 , is virtually insoluble.

- (a) Explain the difference in solubility between MgCl_2 and MgF_2 in terms of lattice energy. [2]

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- (b) Write the balanced chemical equation for the precipitation of MgF_2 from aqueous solutions of MgSO_4 and NaF . [2]

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SECTION C: 2-MARK TARGETED EXAM QUESTIONS (20% TARIFF · Q33–Q40)

33 General Equation for Thermal Decomposition of Group 2 Carbonate — 9701/42/M/J/23/Q8 [2 Marks]

EASY · Syllabus 27.1

Group 2 carbonates undergo endothermic thermal decomposition.

- (a) Write the general equation for the decomposition of MCO_3 , including state symbols. [2]

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34 Trend in Group 2 Hydroxide Solubility — 9701/41/M/J/23/Q8 [2 Marks]

EASY · Syllabus 27.2

The solubility of Group 2 hydroxides varies systematically.

- (a) State the trend in solubility of Group 2 hydroxides down the group from Mg to Ba. [2]

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35 Flame Colour of Calcium Cations — 9701/42/O/N/23/Q8 [2 Marks]

EASY · Syllabus 27.3

Calcium salts produce a distinctive flame colour.

- (a) State the flame colour for Ca^{2+} and name the acid used to clean the wire. [2]

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36 Formula and Medical Role of Barium Sulfate — 9701/41/O/N/23/Q8 [2 Marks]

EASY · Syllabus 27.4

Barium sulfate is used in clinical diagnostics.

- (a) State the chemical formula of barium sulfate and its clinical use. [2]

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37 Decomposition Products of Group 2 Nitrates — 9701/42/M/J/22/Q8 [2 Marks]

EASY · Syllabus 27.1

Group 2 nitrates decompose when heated.

- (a) Name the two gaseous products formed during the decomposition of magnesium nitrate. [2]

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38 Agricultural Role of Calcium Hydroxide — 9701/41/M/J/22/Q8 [2 Marks]

EASY · Syllabus 27.4

Calcium hydroxide is added to farmland.

- (a) State the common name of $\text{Ca}(\text{OH})_2$ and why it is spread on agricultural fields. [2]

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39 Formula of Magnesium Peroxide versus Oxide — 9701/42/O/N/22/Q8 [2 Marks]

EASY · Syllabus 27.3

Oxygen forms different types of binary compounds with metals.

- (a) Give the chemical formulas for magnesium oxide and barium peroxide. [2]

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40 Trend in Sulfate Solubility Down Group 2 — 9701/41/O/N/22/Q8 [2 Marks]

EASY · Syllabus 27.2

The solubility of sulfates changes down Group 2.

- (a) State the trend in solubility of Group 2 sulfates from Mg to Ba. [2]

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SECTION D: HIGH-FREQUENCY CORE REPEATS — 10 MOST FREQUENTLY TESTED QUESTIONS (Q41–Q50)

41 [CORE REPEAT 1] Complete Thermodynamic Analysis of Sulfate Solubility — 9701/42/M/J/23/Q1 [6 Marks]

HARD · Syllabus 27.2

The trend in the solubility of Group 2 sulfates is analysed using enthalpy data.
The trend is shown in Fig. 41.1.

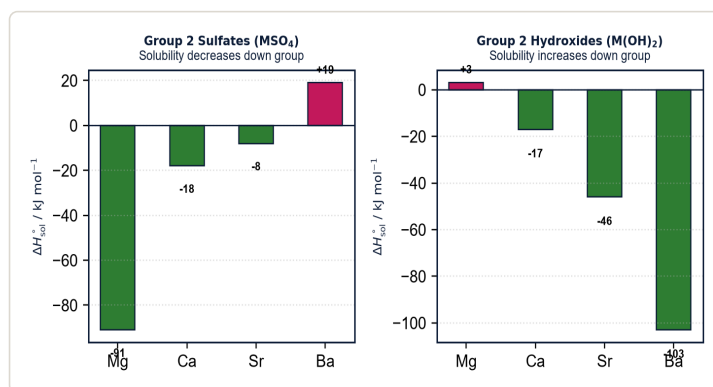


Fig. 41.1: Standard enthalpy of solution trends for Group 2 sulfates.

- (a) State the trend in the solubility of Group 2 sulfates down the group. [1]

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- (b) Explain the trend in terms of the relative changes in lattice energy and hydration enthalpy. [3]

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- (c) Barium sulfate is precipitated by adding sodium sulfate to barium chloride. Write the full and ionic equations. [2]

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42 [CORE REPEAT 2] Carbonate Thermal Stability & Anion Polarisation Mechanism — 9701/41/M/J/23/Q1 [6 Marks] HARD · Syllabus 27.1

The thermal decomposition temperatures of Group 2 carbonates follow the trend in Fig. 42.1.

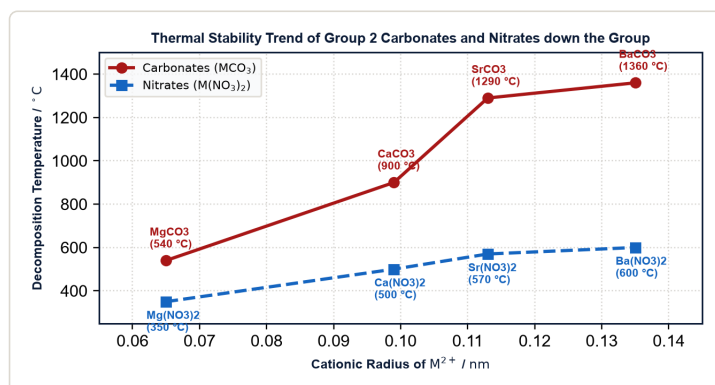


Fig. 42.1: Increasing decomposition temperature of Group 2 carbonates with increasing cation radius.

- (a) State how the decomposition temperature varies with the radius of the M²⁺ cation. [1]

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- (b) Explain why magnesium carbonate decomposes much more easily than barium carbonate. [3]

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- (c) Write balanced equations for the decomposition of CaCO₃ and Ca(NO₃)₂. [2]

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43 [CORE REPEAT 3] Hydroxide Solubility Trend & Industrial Uses — 9701/42/O/N/23/Q1 [6 Marks] HARD · Syllabus 27.2

Group 2 hydroxides become increasingly soluble down the group, reaching an alkaline pH of 13 for Ba(OH)₂.

- (a) Explain why Group 2 hydroxides become more soluble down the group. [3]

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- (b) Describe the use of calcium hydroxide in treating acidic soil and explain the chemical hazard of mixing it with ammonium fertilisers. [2]

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- (c) State the flame colours for strontium and barium compounds. [1]

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44 [CORE REPEAT 4] Nitrate Decomposition Stoichiometry & Gas Volume Calculation — 9701/41/O/N/23/Q1 [6 Marks] · Syllabus 27.1

A sample of 2.12 g of anhydrous strontium nitrate, $\text{Sr}(\text{NO}_3)_2$ ($M_r = 211.6$), is heated strongly until decomposition is complete:
 $2\text{Sr}(\text{NO}_3)_2(\text{s}) \rightarrow 2\text{SrO}(\text{s}) + 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$
 Assume all gas volumes are measured at room temperature and pressure (rtp, molar gas volume = $24.0 \text{ dm}^3 \text{ mol}^{-1}$).

- (a) Calculate the mass of solid residue, SrO ($M_r = 103.6$), formed. [2]

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- (b) Calculate the volume of nitrogen dioxide gas, NO_2 , evolved at rtp. [2]

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- (c) Describe a chemical test to confirm that oxygen gas is also present in the evolved gas mixture. [2]

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45 [CORE REPEAT 5] Polarising Power Definition & Trend Explanation — 9701/42/M/J/22/Q1 [4 Marks] MEDIUM · Syllabus 27.1

The ability of a cation to distort an anion's electron cloud is called its polarising power.

- (a) State two factors that govern the polarising power of a cation. [2]

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- (b) Explain why the polarising power of Group 2 cations decreases down the group from Mg^{2+} to Ba^{2+} . [2]

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46 [CORE REPEAT 6] Reaction of Magnesium with Steam vs Water — 9701/41/M/J/22/Q1 [4 Marks] MEDIUM · Syllabus 27.3

Magnesium reacts differently with cold liquid water and hot gaseous steam.

- (a) Write the balanced chemical equation for the reaction of magnesium with steam. [2]

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- (b) Write the balanced chemical equation for the reaction of magnesium with cold water, and state why the reaction with cold water is extremely slow. [2]

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47 [CORE REPEAT 7] Testing for Sulfate Ions & Solubility of Barium Salts — 9701/42/O/N/22/Q1 [4 Marks] MEDIUM · Syllabus 27.4

A white powder is suspected of containing sulfate ions.

- (a) Describe how to test for sulfate ions in aqueous solution, stating reagents and the positive observation. [2]

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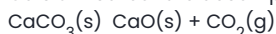
- (b) Write the ionic equation for the precipitation reaction and state why barium sulfate does not dissolve in dilute acid. [2]

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48 [CORE REPEAT 8] Thermal Decomposition of Calcium Carbonate — 9701/41/O/N/22/Q1 [4 Marks] MEDIUM · Syllabus 27.1

Calcium carbonate decomposes on strong heating:



- (a) State the type of reaction and whether it is exothermic or endothermic. [2]

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- (b) Describe a chemical test for the gas evolved, including the observation and balanced chemical equation. [2]

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49 [CORE REPEAT 9] Flame Emission Colour of Barium — 9701/42/M/J/21/Q1 [2 Marks] EASY · Syllabus 27.3

Flame tests identify metal cations.

- (a) State the flame colour for barium and explain why a non-luminous flame must be used. [2]

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50 [CORE REPEAT 10] Relative Basicity of Group 2 Oxides — 9701/41/M/J/21/Q1 [2 Marks] EASY · Syllabus 27.3

Oxides of Group 2 elements are basic.

- (a) State the trend in basicity of Group 2 oxides down the group, and state the oxide with the highest basicity. [2]

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Mark Scheme — Topic 27 — Group 2

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — PAPER 4 MARKING CRITERIA

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
1(a)	Thermal stability increases down Group 2 from MgCO_3 to BaCO_3 [1].	1
1(b)	Down the group, the ionic radius of the M^{2+} cation increases while charge remains constant (+2) [1]; The charge density and polarising power of the M^{2+} cation decreases [1]; Smaller Mg^{2+} causes greater distortion/polarisation of the electron cloud of the CO_3^{2-} ion, weakening the C-O bond and requiring less thermal energy to decompose [1].	3
1(c)	$2\text{Mg}(\text{NO}_3)_2(\text{s}) \rightarrow 2\text{MgO}(\text{s}) + 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$ [1]; $\text{MgCO}_3(\text{s}) \rightarrow \text{MgO}(\text{s}) + \text{CO}_2(\text{g})$ [1].	2
2(a)	$\Delta H^\circ_{\text{sol}} = \Delta H^\circ_{\text{hyd}} - \Delta H^\circ_{\text{latt}}$; For MgSO_4 : $(-1920 + -1041) - (-2870) = -2961 + 2870 = -91 \text{ kJ mol}^{-1}$ [1]; For BaSO_4 : $(-1360 + -1041) - (-2469) = -2401 + 2469 = +68 \text{ kJ mol}^{-1}$ [1].	2
2(b)	Cation radius increases down Group 2, so the distance between the nucleus of M^{2+} and the anion / water lone pair increases [1]; Weaker electrostatic attractions between M^{2+} and the anion / water dipoles decrease the magnitude of both terms [1].	2
2(c)	The sulfate ion is very large compared to M^{2+} , so the inter-ionic distance in MSO_4 is dominated by the sulfate radius and lattice energy decreases only slightly [1]; In contrast, $\Delta H^\circ_{\text{hyd}}$ depends directly on cation radius and decreases steeply, making $\Delta H^\circ_{\text{sol}}$ increasingly positive (endothermic) and reducing solubility [1].	2
3(a)	Moles $\text{Ca}(\text{NO}_3)_2 = 4.92 / 164.1 = 0.0300 \text{ mol}$ [1]; Moles $\text{CaO} = 0.0300 \text{ mol}$ & Δ ; Mass $\text{CaO} = 0.0300 \times 56.1 = 1.68 \text{ g}$ [1].	2
3(b)	Total moles gas = $(4/2 + 1/2) \times 0.0300 = 2.5 \times 0.0300 = 0.0750 \text{ mol}$ [1]; Volume of gas = $0.0750 \times 24.0 = 1.80 \text{ dm}^3$ (1800 cm^3) [1].	2
3(c)	Barium nitrate requires a higher temperature to decompose (higher thermal stability) [1]; Ba^{2+} has a larger radius and lower charge density than Ca^{2+} , polarising the nitrate anion less strongly and weakening the N-O bond to a lesser extent [1].	2
4(a)	Both lattice energy and hydration enthalpy become less exothermic down the group [1]; The OH^- ion is very small, so the sum ($r^+ + r^-$) changes significantly and lattice energy decreases very rapidly down the group [1]; Lattice energy decreases faster than hydration enthalpy, so $\Delta H^\circ_{\text{sol}}$ becomes more exothermic (or less endothermic), increasing solubility [1].	3
4(b)	Calcium: brick-red / orange-red; Strontium: scarlet / crimson; Barium: apple-green [2].	2
4(c)	Thermal energy excites electrons to higher energy levels; when these electrons fall back to lower levels, photons of visible light are emitted with $E = hf$ [1].	1
5(a)	Charge density is the ratio of ionic charge to ionic volume (or surface area) [1]; $(0.099 / 0.065)^3 \approx 1.523^3 \approx 3.53$ & Δ ; Mg^{2+} has approximately 3.5 times higher charge density than Ca^{2+} [1].	2
5(b)	The higher charge density of Mg^{2+} exerts a stronger electric field, more strongly polarising/distorting the electron cloud of the neighbouring CO_3^{2-} ion [1]; This weakens a C-O bond within the carbonate ion, allowing CO_2 to leave at a lower activation energy and lower temperature [1].	2
5(c)	Group 1 cations carry only a +1 charge and have larger radii, giving them much lower charge density and negligible polarising power [1]; Li^+ is an exception because its ionic radius is exceptionally small, giving it a charge density comparable to Mg^{2+} (diagonal relationship) [1].	2
6(a)	Bright white light / white solid powder forms [1]; $\text{Mg}(\text{s}) + \text{H}_2\text{O}(\text{g}) \rightarrow \text{MgO}(\text{s}) + \text{H}_2(\text{g})$ [1].	2
6(b)	Vigorous effervescence / bubbles of gas, cloudy white precipitate / suspension forms [1]; $\text{Ca}(\text{s}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow \text{Ca}(\text{OH})_2(\text{s/aq}) + \text{H}_2(\text{g})$ [1].	2
6(c)	Down the group, atomic radius increases and electron shielding increases [1]; The outermost two s electrons are further from the nucleus and less strongly held, so the sum of ($\text{IE}_1 + \text{IE}_2$) decreases, making oxidation of M to M^{2+} much more favourable [1].	2
7(a)	Neutralises acidic soils to raise soil pH to optimal levels for crop growth [1]; $\text{Ca}(\text{OH})_2$ reacts with NH_4^+ to liberate volatile ammonia gas: $\text{Ca}(\text{OH})_2 + 2\text{NH}_4\text{NO}_3 \rightarrow \text{Ca}(\text{NO}_3)_2 + 2\text{NH}_3(\text{g}) + 2\text{H}_2\text{O}$ [1]; The loss of gaseous NH_3 severely reduces the nitrogen content available to crops [1].	3
7(b)	BaSO_4 has an extremely tiny solubility product K_{sp} (approx. $1.1 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$) [1]; It is completely insoluble in body fluids, so virtually zero Ba^{2+} ions dissolve into the bloodstream [1].	2

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
7(c)	Used as an antacid (milk of magnesia) to neutralise excess stomach acid / treat indigestion [1].	1
8(a)	Mass of CO ₂ lost = 5.00 – 3.90 = 1.10 g [1].	1
8(b)	Moles of CO ₂ lost = 1.10 / 44.0 = 0.0250 mol [1]; Moles of MgCO ₃ decomposed = 0.0250 mol [1]; Mass of MgCO ₃ = 0.0250 × 84.3 = 2.1075 g ⇒ % MgCO ₃ = (2.1075 / 5.00) × 100 = 42.2% [1].	3
8(c)	Ba ²⁺ has a much larger ionic radius and very low charge density [1]; It polarises the CO ₃ ²⁻ anion weakly, so a temperature far above 600 °C (approx. 1360 °C) is required for decomposition [1].	2
9(a)	Enthalpy change when one mole of specified gaseous ions is completely dissolved in water to form an infinitely dilute aqueous solution under standard conditions [2].	2
9(b)	Down the group, the ionic radius of M ²⁺ increases while charge remains +2 [1]; The charge density decreases, so electrostatic attractions between M ²⁺ and the partial negative oxygen atoms of water dipoles become weaker, making H ^o hyd less exothermic [1].	2
9(c)	Mg ²⁺ has a higher ionic charge (+2 vs +1) and a smaller ionic radius (0.065 nm vs 0.095 nm) [1]; Mg ²⁺ has a vastly greater charge density, exerting much stronger ion-dipole attractions on surrounding water molecules [1].	2
10(a)	CaO(s) + H ₂ O(l) → Ca(OH) ₂ (aq/s) [1]; pH ≈ 11 to 12.5 (alkaline) [1].	2
10(b)	MgO has an extremely high melting point (approx. 2850 °C) [1]; Due to very strong electrostatic attractions between small, doubly charged Mg ²⁺ and O ²⁻ ions forming a giant ionic lattice with high lattice energy [1].	2
10(c)	O ²⁻ (s) + 2H ⁺ (aq) → H ₂ O(l) [1]; The oxide ion acts as a Brønsted-Lowry base by accepting protons from H ⁺ ions [1].	2
11(a)	Cycle showing CaCl ₂ (s) → Ca ²⁺ (aq) + 2Cl ⁻ (aq) (H ^o sol) with route via gaseous ions Ca ²⁺ (g) + 2Cl ⁻ (g) (–H ^o latt then ΣH ^o hyd) [2].	2
11(b)	H ^o sol = ΣH ^o hyd – H ^o latt = (–1577 + 2(–378)) – (–2258) [1]; = (–1577 – 756) + 2258 = –2333 + 2258 = –75 kJ mol ^{–1} [1].	2
11(c)	Temperature increases [1]; The dissolution process is exothermic (H ^o sol = –75 kJ mol ^{–1} < 0), releasing heat energy to the aqueous solution [1].	2
12(a)	Li ⁺ and Mg ²⁺ have very similar charge densities (ratio of charge to ionic radius) and electronegativities [2].	2
12(b)	LiNO ₃ decomposes like Group 2: 4LiNO ₃ (s) → 2Li ₂ O(s) + 4NO ₂ (g) + O ₂ (g) [1]; NaNO ₃ decomposes only to nitrite: 2NaNO ₃ (s) → 2NaNO ₂ (s) + O ₂ (g) [1].	2
12(c)	Li ₂ CO ₃ decomposes on heating: Li ₂ CO ₃ (s) → Li ₂ O(s) + CO ₂ (g) [1]; Na ₂ CO ₃ does not decompose at Bunsen temperatures [1].	2
13(a)	Hydration of ions leads to the strict ordering and orientation of water molecules into hydration shells around Ba ²⁺ and SO ₄ ²⁻ [1]; This ordering of solvent molecules creates a large negative entropy change that outweighs the positive entropy of lattice dispersal [1].	2
13(b)	G ^o sol = H ^o sol – TS ^o sol = +19 – (298 × (–0.040)) [1]; = +19 – (–11.92) = +30.9 kJ mol ^{–1} [1].	2
13(c)	Not spontaneous because G ^o sol is large and positive (+30.9 kJ mol ^{–1}) [1]; Because S ^o sol is negative, the –TS term becomes more positive as T increases, making G ^o sol even more positive and keeping BaSO ₄ insoluble [1].	2
14(a)	Labelled diagram showing boiling tube with carbonate clamped horizontally/slanted, Bunsen burner, delivery tube leading into test-tube containing limewater [2].	2
14(b)	1. Distance of Bunsen flame from boiling tube / height of flame [1]; 2. Volume and concentration of limewater [1].	2
14(c)	Order: MgCO ₃ turns cloudy first, then CaCO ₃ , then SrCO ₃ [1]; Suck-back hazard: when heating stops, cooling air in tube contracts and draws cold limewater back into hot boiling tube causing glass to shatter; avoid by removing delivery tube from limewater before turning off flame [1].	2
15(a)	In BaO: –2 [1]; In BaO ₂ : –1 [1].	2
15(b)	The peroxide ion O ₂ ²⁻ is relatively large and unstable in the presence of small, highly polarising cations [1]; Large Ba ²⁺ has low charge density and does not polarise the peroxide ion, stabilising the BaO ₂ lattice; small Mg ²⁺ strongly polarises O ₂ ²⁻ , splitting it into O ₂ ^{•–} and O ₂ [1].	2

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
15(c)	$\text{BaO}_2(\text{s}) + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{BaSO}_4(\text{s}) + \text{H}_2\text{O}_2(\text{aq})$ [1]; Hydrogen peroxide, H_2O_2 (antiseptic / bleach) [1].	2
16(a)	$\Delta H^\circ_{\text{sol}} = \Delta H^\circ_{\text{hyd}} - \Delta H^\circ_{\text{latt}} = (-1360 + 2(-519)) - (-2318)$ [1]; $= (-1360 - 1038) + 2318 = -2398 + 2318 = -80 \text{ kJ mol}^{-1}$ [1].	2
16(b)	The dissolution of $\text{Ba}(\text{OH})_2$ is strongly exothermic ($\Delta H^\circ_{\text{sol}} = -80 \text{ kJ mol}^{-1}$), which makes $\Delta G^\circ_{\text{sol}}$ negative and dissolution highly favourable [1]; For $\text{Mg}(\text{OH})_2$, $\Delta H^\circ_{\text{sol}}$ is endothermic, creating a positive $\Delta G^\circ_{\text{sol}}$ barrier [1].	2
16(c)	Initially turns cloudy/milky due to precipitate of CaCO_3 [1]; With excess CO_2 , precipitate redissolves to form a colourless clear solution of soluble calcium hydrogencarbonate, $\text{Ca}(\text{HCO}_3)_2$ [1].	2
17(a)	Clean a nichrome wire by dipping in concentrated hydrochloric acid and placing in a non-luminous Bunsen flame until no colour is imparted [1]; Dip cleaned wire into sample and hold in the edge of the blue non-luminous flame [1].	2
17(b)	Strontium: scarlet / crimson red [1]; Barium: apple-green [1].	2
18(a)	As cation radius increases, thermal decomposition temperature increases (thermal stability increases) [2].	2
18(b)	Smaller Mg^{2+} has higher charge density and stronger polarising power, distorting the CO_3^{2-} anion and weakening the C-O bond [1]; Larger Ba^{2+} has lower charge density and polarises the anion less, requiring more thermal energy to break the bond [1].	2
19(a)	$\text{Ba}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{BaSO}_4(\text{s})$ [2].	2
19(b)	Dilute HCl reacts with and removes interfering carbonate (CO_3^{2-}) or sulfite (SO_3^{2-}) ions [1]; which would otherwise form insoluble white precipitates (BaCO_3 or BaSO_3) and give a false positive result [1].	2
20(a)	Saturated $\text{Ca}(\text{OH})_2$ has pH ≈ 12.0 to 12.5 ; 0.10 mol dm^{-3} NaOH has pH = 13.0 [1]; NaOH is a stronger base in solution because it is completely soluble and dissociates to give a higher $[\text{OH}^-]$ [1].	2
20(b)	Group 1 cations carry only a +1 charge, so their lattice energies are significantly lower than those of Group 2 hydroxides containing +2 cations [1]; The lower lattice enthalpy makes the dissolution enthalpy $\Delta H^\circ_{\text{sol}}$ more favourable [1].	2
21(a)	1. Brown gas (NO_2) evolved [1]; 2. Glowing splint relights (O_2 evolved) / white solid residue remains [1].	2
21(b)	$2\text{Sr}(\text{NO}_3)_2(\text{s}) \rightarrow 2\text{SrO}(\text{s}) + 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$ [2].	2
22(a)	Ba has a larger atomic radius and more electron shells, resulting in greater electron shielding [1]; The outermost 6s electrons experience a weaker electrostatic pull from the nucleus despite the higher nuclear charge [1].	2
22(b)	The second electron is removed from a positively charged ion (M^+), where there is less electron-electron repulsion and a higher effective nuclear attraction per remaining electron [2].	2
23(a)	$[\text{Ba}^{2+}] = \frac{1.10 \times 10^{-10}}{1.05 \times 10^{-5}} = 1.05 \times 10^{-5} \text{ mol dm}^{-3}$ [1]; This is 10 times lower than the toxic threshold, confirming complete medical safety [1].	2
23(b)	BaCO_3 reacts with stomach acid (hydrochloric acid) to produce soluble and highly toxic BaCl_2 : $\text{BaCO}_3(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{BaCl}_2(\text{aq}) + \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$ [1]; Toxic Ba^{2+} ions would be absorbed into the bloodstream, causing fatal poisoning [1].	2
24(a)	Thermal stability increases down the group from $\text{Mg}(\text{OH})_2$ to $\text{Ba}(\text{OH})_2$ [1].	1
24(b)	Down the group, the ionic radius of M^{2+} increases while charge remains constant (+2) [1]; The charge density and polarising power of the cation decrease [1]; Mg^{2+} polarises the OH^- ion more strongly, weakening the O-H bond and facilitating water elimination at a lower temperature [1].	3
25(a)	Enthalpy change when one mole of an ionic compound is formed from its constituent gaseous ions under standard conditions [2].	2
25(b)	Mg^{2+} has a smaller ionic radius than Ba^{2+} [1]; The sum of the ionic radii ($r^+ + r^-$) is smaller in MgO , resulting in a shorter inter-ionic distance and stronger electrostatic attraction between Mg^{2+} and O^{2-} [1].	2
26(a)	Compound A = CaO (calcium oxide) [1]; Compound B = Ca_3N_2 (calcium nitride) [1].	2
26(b)	Gas C = Ammonia, NH_3 [1]; $\text{Ca}_3\text{N}_2(\text{s}) + 6\text{H}_2\text{O}(\text{l}) \rightarrow 3\text{Ca}(\text{OH})_2(\text{aq/s}) + 2\text{NH}_3(\text{g})$ [1].	2
27(a)	$\text{Mg}(\text{OH})_2(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{MgCl}_2(\text{aq}) + 2\text{H}_2\text{O}(\text{l})$ [2].	2

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
27(b)	Mg(OH) ₂ has very low solubility, providing a mild alkaline buffer that does not burn tissues [1]; Ba(OH) ₂ is much more soluble and releases high concentrations of extremely toxic Ba ²⁺ and caustic OH ⁻ ions [1].	2
28(a)	Thermal decomposition (calcination) in a lime kiln [1]; Heated strongly at approximately 900–1000 °C [1].	2
28(b)	Vigorous exothermic reaction: solid hisses, swells, gives off steam, and crumbles to white powder [1]; CaO(s) + H ₂ O(l) → Ca(OH) ₂ (s) [1].	2
29(a)	Hydration enthalpy H° _{hyd} dominates [1]; Sulfate is a very large anion, so lattice energy changes little down the group; the steep decline in cation hydration enthalpy makes H° _{sol} less exothermic [1].	2
29(b)	Lattice energy H° _{latt} dominates [1]; Hydroxide is a very small anion, so lattice energy decreases much more rapidly than hydration enthalpy, making H° _{sol} more exothermic down the group [1].	2
30(a)	Mass of CO ₂ evolved = 0.500 – 0.270 = 0.230 g [1]; Moles of CO ₂ = 0.230 / 44.0 = 0.005227 mol [1]; Molar mass of MCO ₃ = 0.500 / 0.005227 = 95.65 g mol ⁻¹ ⇒ Ar(M) = 95.65 – 60.0 = 35.65 (or Ar(M) + 16 = 0.270 / 0.005227 = 51.65 ⇒ Ar = 35.7) [1].	3
30(b)	Calcium (Ca, actual Ar = 40.1; within experimental uncertainty for a mixture/hydrated carbonate) [1].	1
31(a)	Calcium reacts much more vigorously [1]; Ca has a larger atomic radius and more electron shielding, meaning its valence electrons are lost more easily (lower IE ₁ + IE ₂) [1].	2
31(b)	Calcium forms sparingly soluble calcium sulfate (CaSO ₄), which precipitates on the surface and slows further reaction [1]; Magnesium forms highly soluble magnesium sulfate (MgSO ₄), leaving a clear solution [1].	2
32(a)	Fluoride (F ⁻) has a much smaller ionic radius than chloride (Cl ⁻), leading to an extraordinarily high lattice energy for MgF ₂ [1]; The hydration enthalpy is insufficient to overcome this massive lattice energy, so H° _{sol} is strongly endothermic [1].	2
32(b)	MgSO ₄ (aq) + 2NaF(aq) → MgF ₂ (s) + Na ₂ SO ₄ (aq) (or Mg ²⁺ (aq) + 2F ⁻ (aq) → MgF ₂ (s)) [2].	2
33(a)	MCO ₃ (s) → MO(s) + CO ₂ (g) [2] (1 mark for correct species, 1 mark for state symbols).	2
34(a)	Solubility increases down the group [1]; from sparingly soluble Mg(OH) ₂ to moderately soluble Ba(OH) ₂ [1].	2
35(a)	Brick-red / orange-red [1]; Concentrated hydrochloric acid, HCl [1].	2
36(a)	BaSO ₄ [1]; Radiocontrast agent / 'barium meal' for X-ray imaging of the digestive system [1].	2
37(a)	Nitrogen dioxide (NO ₂) [1]; Oxygen (O ₂) [1].	2
38(a)	Slaked lime / hydrated lime [1]; To neutralise soil acidity and raise soil pH to promote healthy plant growth [1].	2
39(a)	Magnesium oxide: MgO [1]; Barium peroxide: BaO ₂ [1].	2
40(a)	Solubility decreases down the group [1]; from highly soluble MgSO ₄ to completely insoluble BaSO ₄ [1].	2
41(a)	Solubility decreases down Group 2 from MgSO ₄ to BaSO ₄ [1].	1
41(b)	Both H° _{latt} and H° _{hyd} become less exothermic down the group as cation radius increases [1]; Because the SO ₄ ²⁻ ion is very large, the lattice energy decreases only slightly [1]; The hydration enthalpy of M ²⁺ decreases much more steeply, causing H° _{sol} to become more endothermic (positive), reducing solubility [1].	3
41(c)	Full: BaCl ₂ (aq) + Na ₂ SO ₄ (aq) → BaSO ₄ (s) + 2NaCl(aq) [1]; Ionic: Ba ²⁺ (aq) + SO ₄ ²⁻ (aq) → BaSO ₄ (s) [1].	2
42(a)	Decomposition temperature increases as cation radius increases [1].	1
42(b)	Mg ²⁺ has a much smaller ionic radius and higher charge density than Ba ²⁺ [1]; Mg ²⁺ exerts a stronger polarising power on the electron cloud of the CO ₃ ²⁻ anion [1]; This distorts the carbonate ion and weakens a C–O bond, lowering the activation energy for decomposition [1].	3
42(c)	CaCO ₃ (s) → CaO(s) + CO ₂ (g) [1]; 2Ca(NO ₃) ₂ (s) → 2CaO(s) + 4NO ₂ (g) + O ₂ (g) [1].	2
43(a)	Both H° _{latt} and H° _{hyd} become less exothermic down the group [1]; Because OH ⁻ is a very small anion, the inter-ionic distance (r ⁺ + r ⁻) increases proportionally more, so lattice energy decreases much faster than hydration enthalpy [1]; Consequently, H° _{sol} becomes more exothermic (or less endothermic), favouring dissolution [1].	3

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
43(b)	Neutralises soil acidity [1]; Reacts with ammonium ions liberating volatile ammonia gas: $\text{Ca}(\text{OH})_2 + 2\text{NH}_4^+ \rightarrow \text{Ca}^{2+} + 2\text{NH}_3(\text{g}) + 2\text{H}_2\text{O}$, depleting the soil of vital nitrogen [1].	2
43(c)	Strontium: scarlet / crimson red; Barium: apple-green [1].	1
44(a)	Moles $\text{Sr}(\text{NO}_3)_2 = 2.12 / 211.6 = 0.01002 \text{ mol}$ [1]; Mass $\text{SrO} = 0.01002 \times 103.6 = 1.04 \text{ g}$ [1].	2
44(b)	Moles $\text{NO}_2 = 2 \times 0.01002 = 0.02004 \text{ mol}$ [1]; Volume $\text{NO}_2 = 0.02004 \times 24.0 = 0.481 \text{ dm}^3$ (481 cm^3) [1].	2
44(c)	Insert a glowing splint into the gas mixture [1]; The splint relights / bursts into flame [1].	2
45(a)	1. Ionic charge of the cation [1]; 2. Ionic radius of the cation (or charge density) [1].	2
45(b)	The ionic charge remains constant at +2 [1]; The ionic radius increases down the group, so charge density and electric field strength decrease, weakening polarising power [1].	2
46(a)	$\text{Mg}(\text{s}) + \text{H}_2\text{O}(\text{g}) \rightarrow \text{MgO}(\text{s}) + \text{H}_2(\text{g})$ [2].	2
46(b)	$\text{Mg}(\text{s}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow \text{Mg}(\text{OH})_2(\text{s}) + \text{H}_2(\text{g})$ [1]; $\text{Mg}(\text{OH})_2$ is sparingly soluble and forms an impervious surface barrier layer on the magnesium metal that prevents further contact with water [1].	2
47(a)	Add dilute hydrochloric acid (or dilute nitric acid) followed by aqueous barium chloride (or barium nitrate) [1]; A thick white precipitate forms [1].	2
47(b)	$\text{Ba}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{BaSO}_4(\text{s})$ [1]; BaSO_4 is the salt of a strong acid (H_2SO_4) so the sulfate ion does not react with H^+ to form an undissociated acid [1].	2
48(a)	Thermal decomposition [1]; Endothermic ($\Delta H > 0$) [1].	2
48(b)	Bubble gas through limewater (aqueous calcium hydroxide): limewater turns cloudy / milky [1]; $\text{Ca}(\text{OH})_2(\text{aq}) + \text{CO}_2(\text{g}) \rightarrow \text{CaCO}_3(\text{s}) + \text{H}_2\text{O}(\text{l})$ [1].	2
49(a)	Apple-green [1]; A non-luminous (blue) flame provides high heat and does not mask the emission colour with yellow soot [1].	2
50(a)	Basicity increases down Group 2 [1]; Barium oxide (BaO) has the highest basicity [1].	2